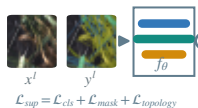


Stage I Supervised warm-up



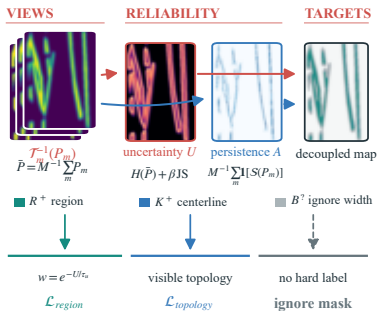
Stage II Semi-supervised training



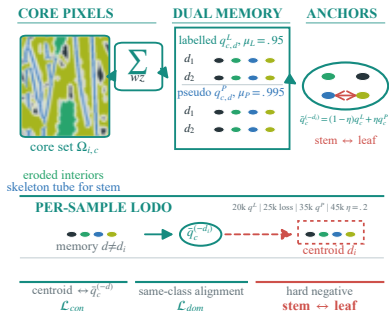
Stage III Frozen inference



1 TRPL Topology-Reliable Pseudo-Labeling



2 TCPM Topology-Core Prototype Memory



3 BAZR Break-Aware Zoom Refinement

